

BARSHINO, Kazakhstan

WMO: 353570

Lat: 49.683N

Lon: 69.517E

Elev: 356

StdP: 97.12

Time Zone: 6.00 (E06)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.9	-31.3	-37.5	0.1	-33.7	-35.0	0.1	-31.1	12.5	-7.9	11.9	-7.9	1.4	180	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	33.9	17.6	32.1	17.3	30.4	16.6	19.2	29.0	18.5	28.4	17.8	27.6	4.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	12.1	21.6	15.3	11.4	21.0	14.3	10.6	20.8	56.1	28.8	53.8	28.3	51.6	27.7	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.5	9.7	8.5	-35.1	36.8	3.4	1.5	-37.6	37.9	-39.6	38.8	-41.5	39.6	-44.0	40.7		
			-35.2	21.3	3.4	1.6	-37.7	22.5	-39.7	23.4	-41.6	24.4	-44.0	25.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.7	-17.3	-16.9	-6.1	7.6	13.9	19.9	21.2	20.2	13.2	4.9	-4.6	-12.9		
	DBStd	15.18	7.72	7.63	6.88	5.25	4.31	3.82	3.52	3.78	4.50	4.98	7.18	8.32		
	HDD10.0	3551	847	754	500	102	14	0	0	0	18	168	437	710		
	HDD18.3	5619	1105	988	757	321	147	28	13	25	164	417	687	968		
	CDD10.0	1250	0	0	1	31	134	297	347	315	114	10	0	0		
	CDD18.3	277	0	0	0	0	9	74	102	81	11	0	0	0		
	CDH23.3	3946	0	0	1	24	227	1058	1316	1117	200	3	0	0		
	CDH26.7	1571	0	0	0	3	44	429	564	463	68	0	0	0		
(14) Wind	WSAvg	3.6	3.4	3.4	4.1	4.1	4.0	3.5	3.6	3.2	3.1	3.3	3.6	3.7		
(15) Precipitation	PrecAvg	234	14	14	12	18	25	37	31	17	13	19	19	15		
	PrecMax	328	28	31	32	40	55	108	77	77	48	55	61	47		
	PrecMin	104	4	4	2	1	3	2	0	0	2	3	3	1		
	PrecStd	64	6	6	9	12	16	31	21	18	12	14	13	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.9	1.1	13.7	26.4	30.2	34.6	36.0	36.2	32.5	22.8	12.1	1.9		
		MCWB	-0.3	-0.2	8.8	13.9	14.7	17.8	19.0	18.3	16.6	12.4	9.7	0.2		
	2%	DB	-0.8	-1.0	7.8	22.9	28.0	32.8	34.0	33.4	28.2	19.4	8.1	0.7		
		MCWB	-1.9	-2.3	5.3	12.3	14.5	18.0	17.7	17.6	15.0	9.9	5.2	-0.7		
	5%	DB	-3.2	-2.5	4.5	19.7	26.0	31.1	32.1	31.1	25.2	16.2	5.7	-0.5		
		MCWB	-4.2	-3.6	2.3	10.6	14.2	16.9	17.5	16.9	13.8	8.5	3.5	-1.7		
	10%	DB	-5.5	-4.7	2.2	16.9	24.0	29.3	30.1	29.2	22.4	13.0	3.6	-2.0		
		MCWB	-6.4	-5.6	0.5	9.6	13.3	16.3	17.0	16.2	12.3	7.2	1.7	-3.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.2	-0.3	9.0	14.5	17.4	21.2	20.3	19.5	18.2	12.5	10.3	0.5		
		MCDB	0.8	0.8	11.8	24.3	24.8	30.4	30.0	29.7	29.6	20.7	11.4	1.4		
	2%	WB	-1.9	-2.1	5.4	13.0	16.0	19.0	19.3	18.6	15.5	10.5	5.7	-0.5		
		MCDB	-0.8	-0.9	7.8	21.1	24.5	28.7	28.8	29.4	26.8	17.1	7.7	0.6		
	5%	WB	-4.1	-3.6	2.2	11.4	15.0	18.1	18.5	17.9	13.9	9.1	3.5	-1.7		
		MCDB	-3.2	-2.5	4.3	18.1	23.4	28.0	28.2	28.4	23.1	14.9	5.3	-0.5		
	10%	WB	-6.4	-5.6	0.6	10.0	14.0	17.3	17.9	17.0	12.7	7.8	1.9	-3.1		
		MCDB	-5.5	-4.7	2.0	15.7	22.9	26.6	27.3	27.4	21.4	12.6	3.6	-2.0		
(35) Mean Daily Temperature Range	MDBR		9.2	10.3	9.2	12.1	14.5	13.9	13.2	13.9	13.9	11.3	8.0	8.7		
	5% DB	MCDBR	7.0	8.1	9.7	17.6	17.6	17.0	16.5	17.5	18.3	17.4	9.4	7.7		
		MCWBR	6.9	7.6	7.0	8.6	7.6	6.4	5.6	6.7	8.3	9.6	6.7	7.0		
	5% WB	MCDBR	7.2	8.1	9.0	15.3	15.4	14.2	13.9	15.2	16.0	15.4	8.4	7.2		
		MCWBR	7.1	7.7	6.8	8.4	7.4	6.2	5.4	6.6	7.8	9.3	6.5	6.7		
(40) Clear-Sky Solar Irradiance	taub		0.296	0.331	0.382	0.413	0.415	0.412	0.429	0.401	0.353	0.342	0.328	0.299		
	taud		2.078	2.001	2.024	2.120	2.182	2.254	2.218	2.281	2.364	2.313	2.189	2.104		
	Ebn at Noon		712	777	805	827	844	848	826	833	841	775	673	646		
	Edh at Noon		85	119	140	142	140	131	134	120	99	87	77	73		
(44) All-Sky Solar Radiation	RadAvg		1.57	2.73	4.00	5.04	6.34	6.68	6.23	5.63	4.29	2.44	1.21	1.07		
	RadStd		0.16	0.17	0.42	0.46	0.43	0.44	0.34	0.42	0.28	0.32	0.14	0.16		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page